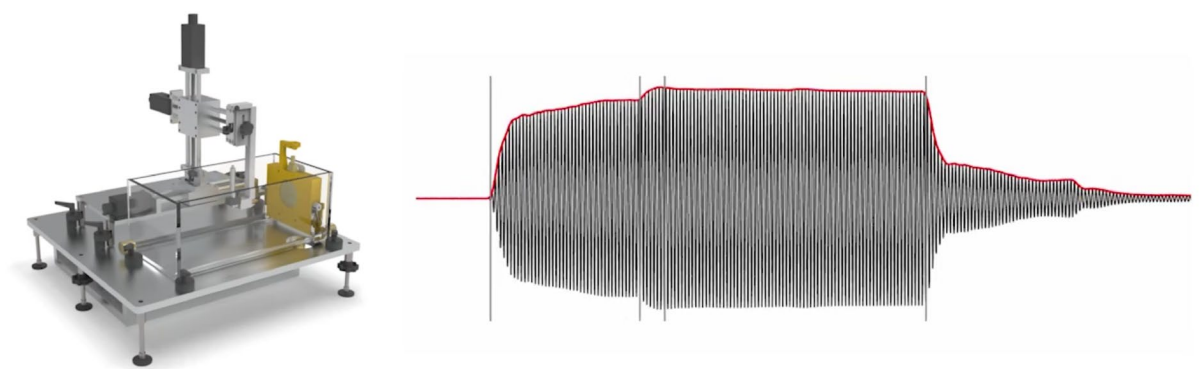


User Manual

SonoRover One Data Analysis



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1 Introduction

1.1 Purpose & Scope

The **SonoRover One Data Analysis** package is a MATLAB-based toolkit designed for advanced post-processing of hydrophone measurements collected using the **SonoRover One** data acquisition system. This manual serves as a comprehensive guide for users who need to convert raw hydrophone data into scientifically meaningful acoustic parameters and visualizations.

The package is specifically tailored for applications in **focused ultrasound metrology**, supporting both research and clinical development environments where precision acoustic characterization is critical.

This toolkit covers the complete data processing pipeline, from raw signal preprocessing to advanced acoustic field analysis and reporting. Its main capabilities include:

- **Preprocessing:** Filtering and conditioning of hydrophone data (e.g., bandpass filtering, Hilbert transform).
- **Pulse Analysis:** Detection of pulse arrival, pulse window selection, and amplitude estimation using multiple methods (FFT, Phasor, Hilbert, and k-Wave simulations).
- **Metric Calculation:** Conversion of voltage signals into **acoustic pressure, intensity, and ISPPA** values, including scaling based on NeuroFUS calibration data.
- **Spatial Filtering:** Suppression of hydrophone reflection artifacts via axial Butterworth filtering.
- **Acoustic Holography:** Full-field sound pressure reconstruction to visualize the acoustic field in 3D space.
- **Visualization and Export:** Generation of plots, comparisons with reference datasets, and exporting of processed results.

The toolkit supports batch processing and is modular, allowing for future extensions to accommodate new metrics or processing techniques.

This manual provides:

- A detailed explanation of the system workflow
- Step-by-step instructions for each processing stage
- Descriptions of each module and function within the package
- Guidance on how to visualize and export results
- Recommendations for troubleshooting and best practices

1.2 Audience

This manual is intended for professionals and researchers who are involved in the measurement, validation, and analysis of focused ultrasound systems. Typical users include:

- **Acoustic Engineers** working on transducer design, beam characterization, and system optimization.
- **Biomedical Researchers** involved in therapeutic ultrasound, neuromodulation, and preclinical validation of ultrasound-based devices.
- **Metrology and QA Specialists** responsible for compliance with acoustic safety guidelines and regulatory standards.
- **Lab Technicians and Operators** using the SonoRover One platform for routine measurements and reporting.
- **Graduate Students** or academic users working on ultrasound-related projects requiring hydrophone data analysis.

A basic understanding of acoustics, hydrophone measurements, and proficiency in **MATLAB** is recommended to effectively use this package.

1.3 System Requirements

To successfully run the **SonoRover One Data Analysis** package, the following software and hardware prerequisites are recommended:

1.4 Software Requirements

- **MATLAB** R2024a or later
(First official MATLAB release of 2024, required for compatibility with object-oriented features and toolboxes used)
- Required MATLAB Toolboxes:
 - **Signal Processing Toolbox**
 - **Image Processing Toolbox** (for holography visualization)
- **k-Wave Toolbox** (open-source MATLAB toolbox for acoustic wave field simulations)
- **natsortFiles** (MATLAB utility for natural sorting of filenames)
- **crossing** (utility function for zero-crossing detection)

1.5 Hardware Recommendations

- **Processor:** Multi-core CPU (Intel i5/i7/AMD Ryzen 5 or higher)
- **Memory:** Minimum 16 GB RAM (32 GB recommended for large holography reconstructions)
- **Storage:** SSD drive recommended for fast data read/write (minimum 1 GB free space for datasets and outputs)
- **GPU:** Optional, CUDA-compatible GPU if you plan to extend functionality with GPU-accelerated MATLAB functions

Platform

- Compatible with **Windows** (tested on Windows 11)
- **macOS** and **Linux** may require minor adjustments for file paths and toolboxes.

1.6 Disclaimer

This user manual and accompanying documentation are provided for informational purposes only and pertain specifically to the in-house developed metrology system described herein. While every effort has been made to ensure accuracy, completeness, and clarity in instructions, procedures, and safety guidelines, the creators and distributors make no guarantees or warranties of any kind—expressed or implied—regarding the accuracy, reliability, or suitability of the provided information for any particular application.

Users are solely responsible for the proper setup, calibration, operation, and maintenance of the metrology system as detailed within this manual. The creators and distributors shall not be held liable for any damage, injury, or loss resulting directly or indirectly from improper installation, misuse, unauthorized modification, or deviation from the recommended procedures outlined in this manual.

The user acknowledges that the performance of the metrology system may vary based on environmental conditions, handling practices, and specific use-case scenarios. Users should undertake regular calibration and verification procedures to ensure optimal performance and reliability.

All safety instructions and best practices must be followed at all times. By operating this metrology system, the user accepts full responsibility for compliance with all applicable safety standards, regulations, and procedures.

2 Installation

2.1 Download and add package to MATLAB path

Go to GitHub repository page: <https://github.com/Donders-Institute/Radboud-FUS-measurement-kit/tree/release/SonoRover%20One%20Data%20Analysis> Click "code" > Download ZIP and extract to an desired working directory.

2.2 Folder structure overview and dependencies

- SonoRover-One-Data-Analysis/
 - |— classes/ # Object-oriented classes for data processing
 - | |— dataPrep.m
 - | |— dataCalc.m
 - | |— dataVis.m
 - | |— dataExp.m
 - | |— readNF.m
 - |— externalData/ # Contains calibration data and supporting NeuroFUS files
 - | |— HydrophoneType # Hydrophone calibration data
 - | |— NFD.mat # Reference data of NeuroFus portfolio
 - |— functions/ # Utility functions
 - | |— cwater.m
 - | |— readIniFile.m
 - |— SonoRover One # output example files
 - |— Toolboxes/ # Third-party toolboxes
 - | |— crossing.m # downloaded form MATLAB central
 - | |— natsortfiles.m # downloaded form MATLAB central
 - | |— k-Wave # downloaded from <https://www.k-wave.org/>
 - |— index.m # Main script file (entry point)

2.3 First-Time Setup

- Add the correct hydrophone txt calibration data in externalData folder. The data is read at the initialisation of an dataCalc.m object.

```
% Initialize hydrophone structures for different devices
hydrophone(1).fileName{1} = 'HGL0200-2845_AG2010-1378-20_CA_20230511.txt';
hydrophone(1).Type = 'Hydrophone HGL 0200 SN2845';
hydrophone(1).SN = '2845';
hydrophone(1).calDate{1} = datetime('2023-05-11');
hydrophone(1).sensitivity{1} = obj.readCalData(hydrophone(1).fileName{1});

% Recalibration on 26-09-2024
hydrophone(1).fileName{2} = 'HGL0200-2845_AG2010-1378-20_CA_20240826.txt';
hydrophone(1).calDate{2} = datetime('2024-08-26');
hydrophone(1).sensitivity{2} = obj.readCalData(hydrophone(1).fileName{2});
```

- Building the NeuroFus data mat file using the provided .xls reference data sheets by Sonic Concepts by defining and correctly referencing using the readNF.m class/

```

%***** 2-CHANNELS *****
%CTX-500-006 2-Ch. Focal Steering - Verification - RMA2230209-1
obj.NFdata(1,1).fileName = 'CTX-500-006 2-Ch. Focal Steering - Verification - RMA2230209-1 - TPO-105-010';
obj.NFdata(1,1).dataArea = 'N19:X80';
obj.NFdata(1,1).propertiesArea = 'R5:T16';

obj.NFdata(1,2).fileName = 'CTX-500-006 2-Ch. Focal Steering - Verification - RMA2230209-1 - TPO-203-035';
obj.NFdata(1,2).dataArea = 'K19:R80';
obj.NFdata(1,2).propertiesArea = 'O5:Q16';

% CTX-250-009 2-Ch. Focal Steering - Verification - RMA2230209-1
obj.NFdata(2,1).fileName = 'CTX-250-009 2-Ch. Focal Steering - Verification - RMA2230209-1 - TPO-105-010';
obj.NFdata(2,1).dataArea = 'P19:AB82';
obj.NFdata(2,1).propertiesArea = 'T5:V16';

```

- Open index.m which is the main file and define parameters in the following sections;
 - **Initialize environment** make sure you set the correct paths
 - pr.defaultDataPath
 - pr.defaultDataPathTesting -> omits the input dialog box and the browsing actions
 - pr. neuroFUSFolderLocation -> location of the .xls reference sheets
 - **Define input parameters**
 - pr.rereadNFD = true -> make sure to build the MATLAB data file in case it does not exist or needs updating after adding a new reference file

3 Workflow Overview

3.1 SonoRover One acquisition files

Data is collected using the SonoRover One system which is described in a separate user manual which contains the following actions

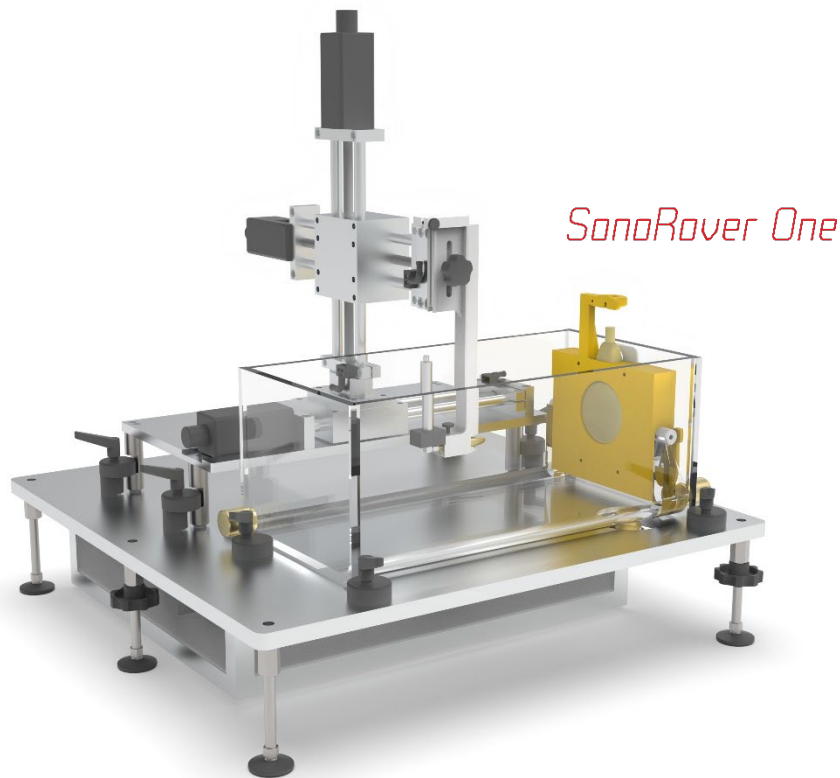


Figure 1. The SonoRover One metrology setup

- Mechanical Probing and Stage-Alignment
- Water Preparation
- Transducer Mounting
- Transducer Connection
- Hydrophone Mounting.
- Acoustical axis estimation
- Define & execute measurement sequences

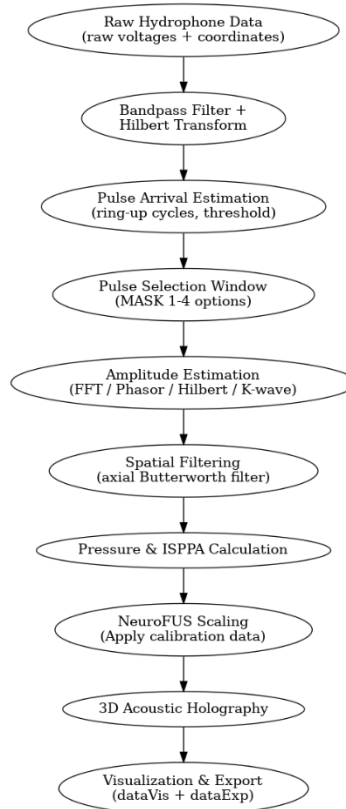
Output example files are provided in the SonoRover One folder for each sequence consist of

- *data.acd # Legacy file, not used
- *data.ini # text file with SonoRover One acquisition properties
 - Versions
 - Equipment
 - Sequence
 - Grid

- Acquisition
- ACD processing
- *data.raw # Raw data
- *data.csv # Absolute and relative measurement position coordinates

3.2 Post-processing steps:

1. Data import & preprocessing
2. Pulse selection and windowing
3. Amplitude estimation
4. Spatial filtering & reflection mitigation
5. Calculation of acoustic metrics (Pressure, ISPPA)
6. Holography reconstruction if applicable
7. Visualization & Export



4 Step-by-Step Guide

4.1 Initialization

Read and prepare SonoRover One data by creating an object prepData.

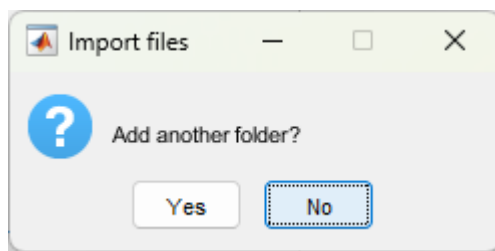
- Class **dataPrep.m** [prepData, setNrs] = dataPrep(pr,p1);

Handles the import and preprocessing of raw hydrophone data. Steps include:

- Reading raw data and coordinates
- Applying bandpass filtering
- Envelope detection (Hilbert transform)
- Estimating pulse arrival and window selection

1. When the devMode parameter is false, a dialog box appear, navigate to the folder which contains output files by the SonoRover One (See paragraph 3.1). This will read all sequence files.

When files need to be appended select "Yes". This will read the additional sequence files alongside the first import. If finished continue by pressing "No"



2. A finite impulse response (FIR) filter with a fixed bandwidth of 100 kHz around the center frequency and a relatively high filter order of 100 was applied. By using the `filtfilt` function, zero-phase filtering was achieved, which is essential for accurately estimating the arrival time and determining the distance from the hydrophone to the exit plane.
3. The envelope was calculated using the Hilbert transform and normalized for pulse index estimation and optional ring-up characteristics and additional analyses.
4. Estimate the arrival of the pulse at the hydrophone and corresponding selection window using the estimated pulse index based on thresholding and based on axial hydrophone location w.r.t. the transducer.
5. `prepData.pulseIndexEst(setNrs,1,ringUpcycles,selectionCycles,time,threshold,EPoffset);`

- | | |
|---------------------|--|
| a. setNrs: | the sequence number selection |
| b. ringUpcycles: | number of periods to exclude from the onset of the pulse, (skipping ring up phase) |
| c. selectionCycles: | the number of periods to take into account for amplitude estimation |
| d. time: | a certain timeperiod [microseconds] from the start of pulse |
| e. threshold: | [%] of normaliseer amplitude |

- f. EPOffset the length between the center of the membrane and the exit plane (specific for each transducer type)

Figure 2 show the estimation of the pulse arrival sample versus the axial distance w.r.t. the transducers radiating surface using the thresholding method in blue and the linear propagating soundwave characteristics in red.

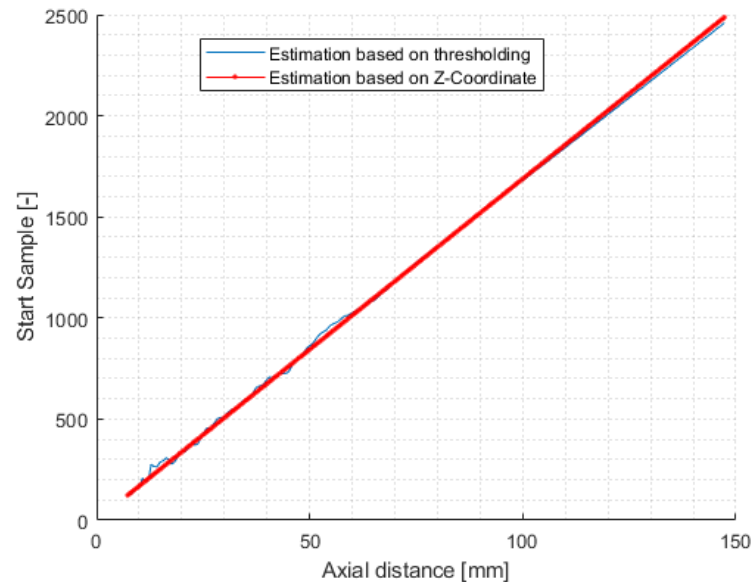


Figure 2. Onset of pulse detection versus Axial distance

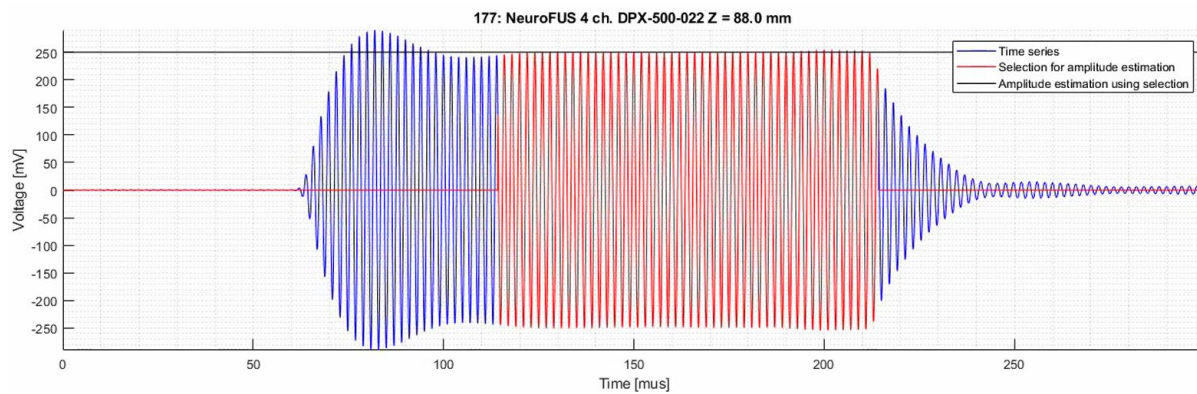
6. Estimate the pulse selection window used as the signal input for amplitude estimation.

```
prepData.pulseWindow(setNrs,mask);
```

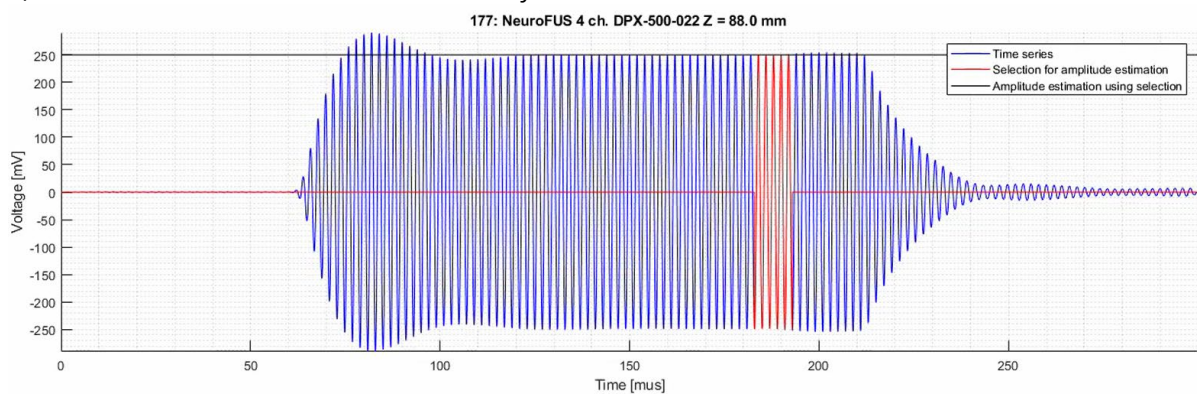
- a. setNrs: the sequence number selection
- b. mask
 - 1: in between end of ring up period and end of pulse
 - 2: in between 3TOF minus selection Cycles and 3TOF
 - 3: in between end of ring up period and end of ring up period + selection cycles
 - 4: at a certain time selection – selection Cycles to certain time.

For each mask type the selection window will be visualized in the figure 3a-d using 25 ringup cycles, 5 selection cycles, 80 microseconds time, threshold of 10% and a offset of 7.3 mm.

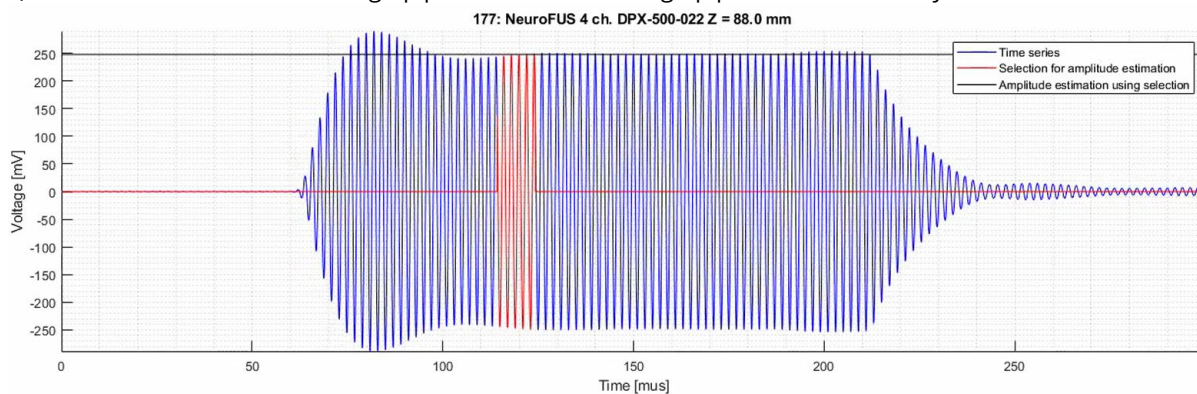
- a) **Mask 1** : in between end of ring up period and end of pulse



b) **Mask 2:** in between 3TOF minus selection cycles and 3TOF.



c) **Mask 3:** in between end of ring up period and end of ring up period + selection cycles



d) **Mask 4:** at a certain time selection – selection cycles to certain time

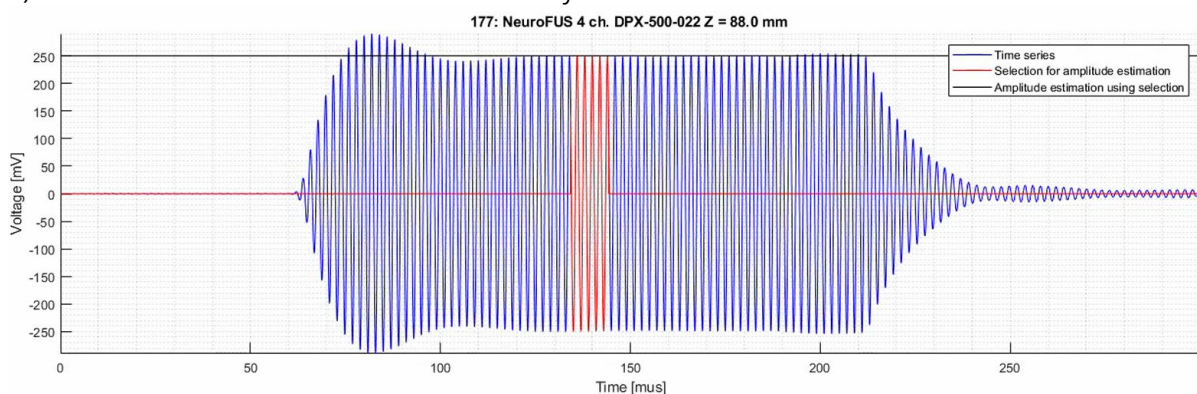


Figure 3 a-d. Visualization of the different selection schemes for amplitude estimation.

4.2 Data processing

- Class **dataCalc.m** `calcData = dataCalc(prepData);`

Core calculation engine which performs:

- Amplitude estimation (FFT, Phasor, Hilbert, K-wave)
- Spatial filtering (Butterworth)
- Pressure and intensity calculations
- ISPPA computation and scaling with NeuroFUS data
- Acoustic holography (3D sound field reconstruction)

1. Creates the object with the `prepData` as input data object. In the constructor of the file hydrophone calibration data is set accordingly:

```
% Initialize hydrophone structures for different devices
hydrophone(1).fileName{1} = 'HGL0200-2845_AG2010-1378-20_CA_20230511.txt';
hydrophone(1).Type = 'Hydrophone HGL 0200 SN2845';
hydrophone(1).SN = '2845';
hydrophone(1).calDate{1} = datetime('2023-05-11');
hydrophone(1).sensitivity{1} = obj.readCalData(hydrophone(1).fileName{1});

% Recalibration on 26-09-2024
hydrophone(1).fileName{2} = 'HGL0200-2845_AG2010-1378-20_CA_20240826.txt';
hydrophone(1).calDate{2} = datetime('2024-08-26');
hydrophone(1).sensitivity{2} = obj.readCalData(hydrophone(1).fileName{2});

% Add a placeholder date for the next calibration
hydrophone(1).calDate{end+1} = datetime('3000-01-01');
```

2. Estimation of the amplitude using the selected data window

```
calcData = calcData.calcAmplitude(ampEstMethod);
```

- a. `ampEstMethod`
 1. FFT based
 2. Phasor based
 3. Hilbert transform
 4. using K-Wave toolbox

3. In case of an axial acquired profile in which only movement along the z axes occurs spatial filtering is applied to mitigate the possible reflection interference from the hydrophone itself which happens at the 3 TOF

```
calcData = calcData.calcSpatialFiltering(scf,ripple,passBandAtten);
```

- a. `scf` spatial cutoff frequency [1/mm]
- b. `ripple` passband ripple [dB]
- c. `passbandAtten` stopband attenuation in [dB]

4. The next steps convert the raw hydrophone electrical signal from Volts to pressure

```
calcData = calcData.calcPressure();
```

For this, hydrophone calibration data defined in the constructor of the class upon object creation in step 1 was used. In case of multiple calibration dates of a single hydrophone, figure depicts both sensitivity curves.

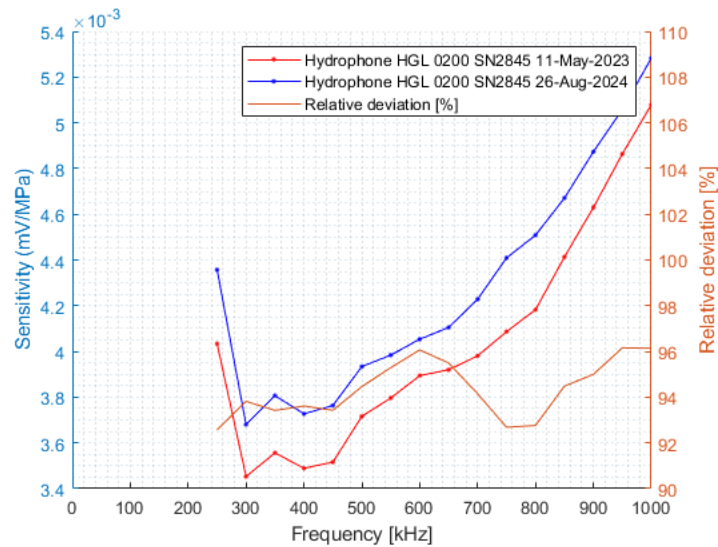


Figure 4. Hydrophone sensitivity curves of multiple calibration dates

5. Consequently, the intensity and ISPPA are calculate using the following functions

```
calcData = calcData.calcIntensity();
calcData = calcData.calcISPPA();
```

6. In case of the comparison with reference data provided by Sonic Concept of the NeuroFus transducers, the NFD structure constructed in the import and prepare data section (paragraph 2.3) was used to scale the calculated ISPPA profiles using the ratio of the max power per channel setpoints of the NeuroFus and SonoRover One profiles.

```
calcData = calcData.calcISPPA2NFscale(NFD);
```

7. In case a cross-sectional plane in front of the transducer was measure. Holography using the K-Wave toolbox functionality can be performed.

```
calcData = calcData.calcHolography(ipf,zv);
```

- a. ipf [#] Interpolation factor of amplitude and phase data
- b. zv [m] The axial vector used for reconstruction of the 3D volume data

4.3 Visualisation

- Class **dataVis.m** dataVisual = dataVis(prepData,calcData);

Visualization toolbox. Generates:

- Pulse selection plots
- Axial profile comparisons with NeuroFUS data
- 3D holography visualizations

Upon creation of the object, a folder “postprocessing” has been created in which all the visualization from all methods are stored

1. To visualize the filtered pulse selection recording for each sequence the following method can be used. This will generate a video of all pulses within a sequence for all sequences. A snapshot of a possible pulse visualisation is depicted in figure 3a-d.

```
dataVisual.pulseSelection(setNrs,xAxis{2},yAxis{1});
```

- | | |
|-----------|---|
| a. X axis | 1: Samples [#]
2: Time [μ s]
3: Cycles [#] |
| b. Y axis | 1: Voltage [mV]
2: Pressure [MPa] |

- To visualize the axial profiles for all sequences in a single graph or in a container in which you can scroll trough the single profiles the following method can be used.

```
dataVisual.axialProfiles(setNrs,xAxis{1},yAxis{y},NFD,singleView{sv});
```

- | | |
|---------------|---|
| a. setNrs | For selected sequences |
| b. X axis | 1: Distance WRT exit plane [mm] |
| c. Y axis | 1: Voltage [mV]
2: Raw & Filt pressure [MPa]
3: Pressure [MPa]
4: ISPPA [W/cm^2]
5: ISPPA scaled [W/cm^2] |
| d. singleView | 1: true: all-in one figure like displayed in figure 5
2: false: embedded in a container to scroll through the profiles |

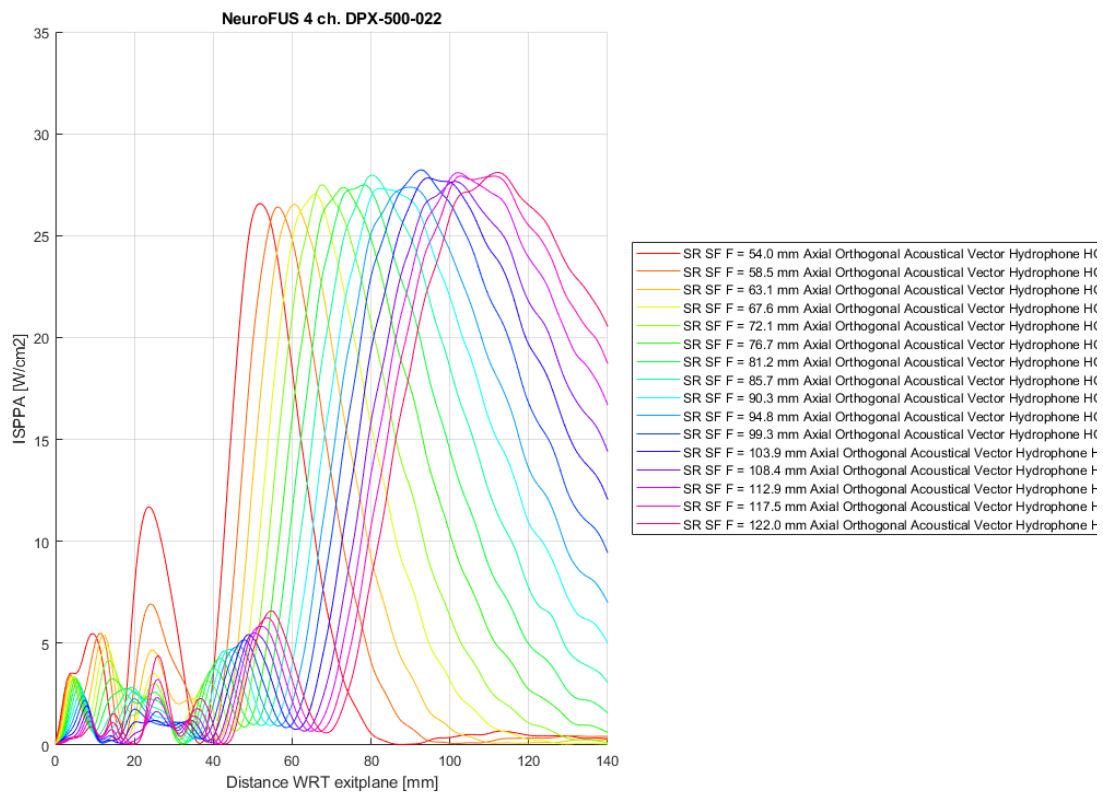


Figure 5. Visualization of ISPPA profiles for specific focus setting

- To visualize the results obtained using the holography method, the following visualization method can be used

`dataVisual.holography(transverseLoc,zv)`

- `transverseLoc` Axial location vector in [mm] of cross-sectional views
- `zv` Axial vector for cross-sectional holography calculations, see paragraph 4.2.7

1. A Native versus Holography derived central axis profiles are plotted to compare the actual measured axial profile with the reconstructed one based on the holography calculation.

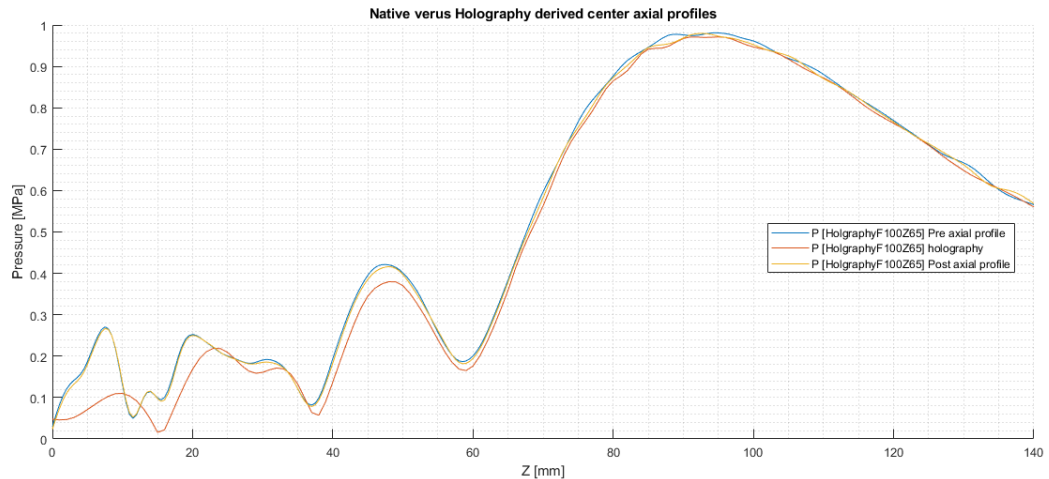


Figure 6. Visualization of the axial profiles

2. The raw amplitude and phase obtained at the single cross-section used as input for the holography method. The input plane has been cropped to maintain the circle symmetry nature of the pressure field. Finally an FFT based interpolation in the horizontal and vertical direction are performed before feeding into the holography algorithm.

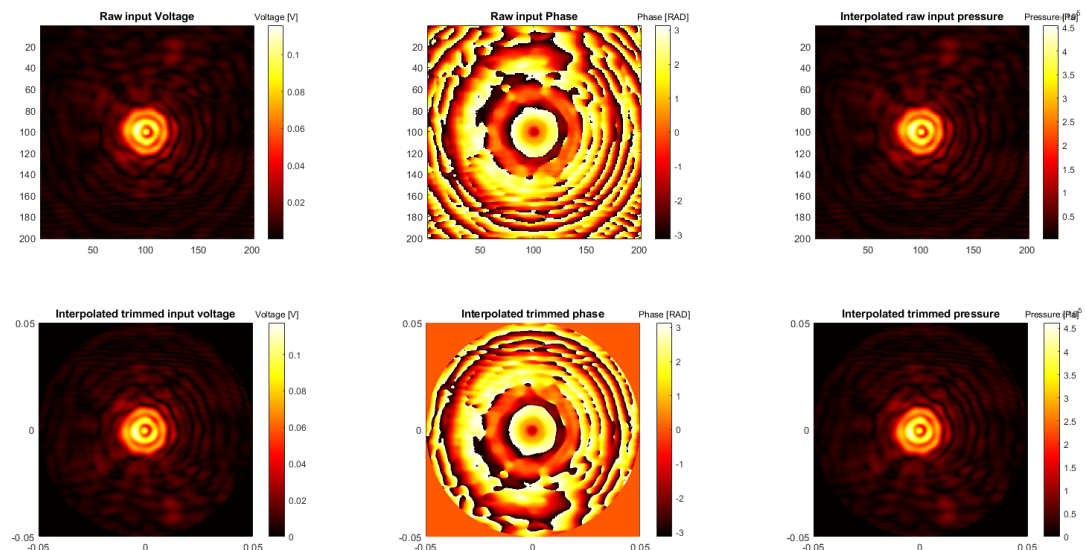


Figure 7. Visualization of the cross-sectional views of the input amplitude and phase and its masking and interpolation.

3. Cross-sectional views defined by `transverseLoc` are displayed

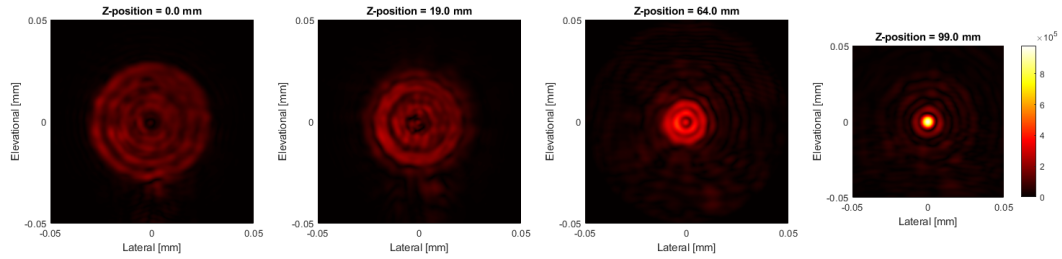


Figure 8. Visualization of the pressure at different axial cross-sections as a result of the angular spectrum algorithm.

4. Lateral, elevational and Transversal cross-sections of the 3D pressure field are viewed

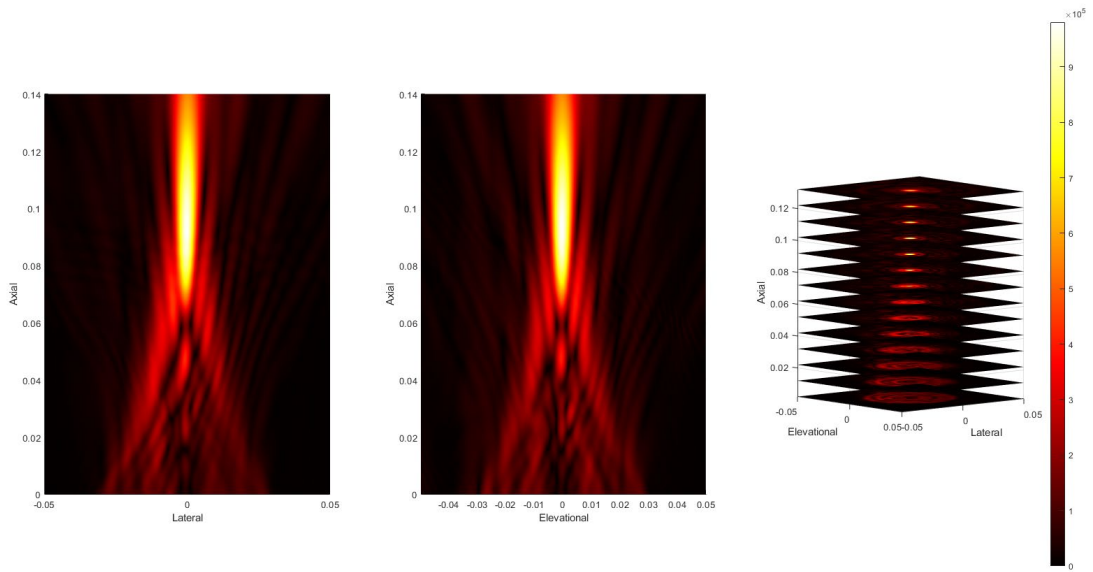


Figure 9. Visualization of the lateral and elevational planes supplemented with a sliced 3D view of the pressure.

4.4 Data Export

- Class **dataExp.m** `expData = dataExp(prepData, calcData);`

Upon creating the `dataExp` object, a designated folder named `Export` will be created in the location of the input files.

1. The export of the pressure and the ISPPA data alongside measurement properties stored in a struct as depicted below.

`expData.press_ISPPA(fileName)`

- a. `fileName` Custom filename of the export structure

```
>> data{1}

ans =

    struct with fields:

        exportDate: 26-Mar-2025 14:07:46
        metrologySetup: 'SonoRover One'
            type: 'Axial Optimal Acoustical Vector'
            amp: 'NeuroFUS 1 x 4 ch. or 1 x 2 ch. TPO junior 203-035'
            power: 4.8590
        powerUnit: 'W'
        transducer: 'NeuroFUS 4 ch. DPX-500-022'
            focus: 54
        focusUnit: 'mm'
            x: [281x1 double]
            y: [281x1 double]
            z: [281x1 double]
        corUnit: 'mm'
        pressure: [1x1 struct]
        ISPPA: [1x1 struct]
```

5 Changelog

5.1 Version history

March 2025	Initial version

6 Contact / Support

For direct contact: fus@ru.nl

6.1 Links to repository or project page (e.g., GitHub)

Code resources	https://github.com/Donders-Institute/Radboud-FUS-measurement-kit/tree/release/SonoRover%20One%20Data%20Analysis